CSE 516 - PROBABILISTIC GRAPHICAL MODELS

MAJOR PROJECT

MILESTONE 1 - PROPOSAL AND SCOPING

Deep End-to-End Causal Inference:

Bayesian Structure Learning and Simulation-Based Treatment Effect Estimation under Heterogeneous, Partially Observed Data

Track: Theoretical AI

Group: G5_TH_PGM

Project Repository: github.com/masoomchoksi/G5_TH_PGM

Base Implementation: Microsoft Causica (microsoft/causica)

M1 focus: Reducing and analysing the boundary conditions of DECI Theorem 1, and testing it in an empirical way, but doing so under controlled conditions.

1. Project Definition and Track

This is a Theoretical AI project. The project commits to one principal methodological question: the validity and boundary of the graph-recovery guarantee stated as Theorem 1 in Deep End-to-end Causal Inference (DECI). The empirical component is supporting evidence for the theoretical analysis, not a substitute for it.

M1 requirement	Project commitment
Selected algorithm / property	DECI causal graph-learning framework; principal focus on Theorem 1 graph-recovery guarantee.
Mathematical question	Under exactly which assumptions does DECI's ELBO optimum recover the true causal graph, and where does the proof cease to apply?

Benchmark plan	Synthetic causal graphs and controlled perturbations, with semi-synthetic causal-effect benchmarks for supporting ATE/CATE analysis.
Primary baseline	NOTEARS-MLP for nonlinear graph-recovery comparison; PC and ICALiNGAM as complementary classical comparators.
SOTA anchor	Vashishtha et al., “Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference,” ICLR 2025, with Microsoft Research affiliation.

2. Project Summary

The problems of causal inference involve answering questions about changes in outcomes under hypothetical interventions. An unobserved causal structure makes simultaneous identification of graph topology and the performance of intervention-aware inference a fundamental problem. DECI addresses this by factoring it all into a single probabilistic model, thereby combining three major tasks – Bayesian graph discovery, nonlinear structural function estimation and simulation-driven treatment effect calculation.

This project looks at the graph-recovery guarantee in Theorem 1 and does not simply reproduce the empirical benchmarks. Specifically, we re-derive the mathematical progression between maximum likelihood estimation, ELBO optimization, KL regularization and the DAG-penalized prior. In addition, we identify the precise proof steps that no longer hold when there is mixed-type data and amortized imputation of missing values, and consider whether these theoretical bounds are reflected in empirical studies of controlled experiments.

3. Problem Statement, Motivation and Significance

1. Problem statement

In conventional causal pipelines, causal discovery and causal inference are usually kept apart. A graph is obtained from observational data first and then kept constant while computing interventions and treatment effects. The downside is that if the graph is wrong, then the further causal-effect estimates would carry the structural error. DECI, on the other hand, does not assume any certainty about which causal graph is true but instead couples graph learning with structural-function learning and intervention simulation.

The problem studied here is not whether DECI can produce good benchmark scores in general. The focused question is whether the formal recovery guarantee can be reconstructed from its stated assumptions and, if an assumption is relaxed, what exact step of the argument fails.

2. Motivation

It is vital to have precise structural assumptions when carrying out causal discovery and causal effect estimation. Nevertheless, a high empirical score does not imply that the theorem can be applied to the particular process generating the data. DECI is an excellent example of end-to-end PGM because it integrates the three aspects of graph representation, learning, and causing of intervention. The issues of mixed types and missing values point to the fact that they are of little theoretical relevance, since they are out of range of applicability of the continuous additive noise theorem.

3. Significance

The expected value of the project is a theory-to-practice map: each theorem assumption will be linked to a model component, a proof step, and an empirical diagnostic. This makes the project more than an implementation exercise and directly answers the course question of where the PGM reasoning is.

4. Research Questions and Hypothesis

1. Can DECI Theorem 1 be rigorously re-derived from its stated assumptions?

2. In the limit of infinite data, what does maximum likelihood recovery of the ground-truth joint distribution contribute to the graph-recovery argument?

3.What is the relation between the convergence of the ELBO-maximizing variational posterior and the MLE solution, and what do the KL term and DAG-constrained prior do?

4. What specific conditions of the theorem are not satisfied by mixed-type/discrete variables and what step of the proof does not make sense in this case?

5. What is the reason that the missing-data extension (based on an approximate amortized imputation posterior) does not belong to the original proof?

6. How do the theoretical limits materialize in actual practice of graph recovery and estimation of ATE/CATE?

Working hypothesis

As the sample size increases, if the continuous additive-noise, identifiability, no-latent-confounding, correct-specification and regularity conditions of the theorem hold, then the behavior of DECI should improve in terms of graph recovery. If these assumptions are not met, posterior uncertainty should grow as the structural error grows and ATE/CATE error should diminish further downstream. This is a suggestion, not an M1 result.

5. Mathematical Formulation

1-Causal graph

Assume that the causal structure is a directed acyclic graph (DAG) G = (V, E), with each node as a random variable and a directed edge Xi → Xj as a direct causal dependency between random variables Xi and Xj.

2-Structural equation model

For each variable Xi, the nonlinear additive-noise SEM is:

$$Xi= fi(Xpa(i)) + Zi$$

Here Xpa(i) is the parent set of Xi, fi is a nonlinear structural function, and Zi is an exogenous noise variable. The project will explicitly track which theorem assumptions are required for identifiability of the causal structure.

3 Factorization and conditional independence

If the graph is a DAG, the joint distribution can be decomposed along the graph (the DAG Markov assumption):

$p(x1,),xd \mid G,)=ip(xi \mid xpa(i), i)$

Conditional-independence statements are also represented by the graph using d-separation. Thus, the absence of an edge is not only visual, but also represents conditional independencies that are implied by the graph, together with the Markov assumptions. The project will explore the impact of misidentified and/or non-identified structures on these implied dependencies.4-Bayesian structure learning

DECI does not consider G as fixed, but rather as uncertain. The target is a posterior $q\varphi(G)$ over a DAG, which is obtained by optimizing a variational objective that combines a DAG-prior, a KL term and the data likelihood. In the large data limit, the project will build the relationship between ELBO objective and MLE solution.

5-Intervention and causal effect

For treatment XT and outcome XY, the intervention do(XT = a) replaces the treatment-generating mechanism with the chosen value. The Average Treatment Effect is:

$$ATE(a,b)=E[XY|do(XT=a)]-E[XY|do(XT=b)]$$

CATE controls for relevant covariates. The quantities are estimated by DECI using simulated versions of the learned structural equations when the graph is intervened, while taking into account uncertainty of candidate graphs.

6. Mandatory PGM Mapping: Representation, Learning and Inference

Vertical	What the project represents / computes	Why appropriate
Representation	Random variables Xi; DAG nodes; directed causal edges; parent sets; nonlinear SEMs; exogenous noise; graph posterior $q_\varphi(G)$.	A DAG makes causal dependencies and conditional-independence assumptions explicit and supports intervention semantics.
Learning	Using Bayesian structure learning combined with functional/parameter learning. ELBO can be optimized by employing a DAG-constrained prior to learn nonlinear structural mechanisms and noise distributions.	Because the underlying causal graph is unobserved, joint Bayesian structure learning is required alongside the continuous estimation of structural parameters and non-linear mechanisms.
Inference	Posterior over graphs → intervention do(XT=a) → structural simulation → outcome distribution → ATE/CATE; uncertainty is marginalized over graph samples.	Exact inference is generally impractical for nonlinear flow-based mechanisms and graph uncertainty; simulation provides an intervention-aware route to causal quantities.

1. Conditional-independence assumptions

- The narrative opens with the process of mandatory PGM mapping, which involves the representation, learning and inference of PGM.
- It is a Markov distribution with respect to the DAG, therefore "d-separation" is a conditional independence in the model assumptions.

- The theorem setting assumes no latent confounders—this is not a realistic assumption, but the project will separate out that one from more realistic partially observed experiments.

2. Representation → Learning → Inference chain

Observational data – Bayesian DAG posterior – nonlinear SEM learning – intervention simulation – ATE/CATE.

7.Theoretical Focus: DECI Theorem 1

The main theoretical problem is to reproduce the proof, not to state the theorem as a "black box. The analysis will be presented in the following manner.

Proof component	Question to be answered	Planned validation
MLE recovery	Under the model and identifiability assumptions, why does maximum likelihood recover the ground-truth joint distribution and causal graph in the infinite-data limit?	Formal derivation and controlled increasing-sample experiments.
ELBO-to-MLE convergence	Why does an optimal variational solution approach the MLE solution as data grow?	Track each optimization/objective term and reproduce the supporting convergence argument.
KL and DAG prior	How does the KL regularization and differentiable DAG-constrained prior affect the variational posterior over graphs?	Ablation/diagnostic analysis where computationally feasible.
Boundary: mixed type	Which continuous additive-noise/identifiability step no longer transfers directly to mixed discrete-continuous variables?	Compare continuous and mixed-type settings.
Boundary: missing data	How does approximate amortized imputation change the objective and invalidate a direct application of the original proof?	Vary missingness rate and report graph/ATE-CATE degradation.

1 Theorem assumptions to track

Assumption	Role	Failure mode to study
Correct specification	True data-generating process lies in the model family.	Model mismatch can prevent exact recovery.
No latent confounding / causal sufficiency	Relevant common causes are observed.	Unobserved confounding can make the learned graph ambiguous.
Structural identifiability	Distribution identifies a unique causal structure in the stated class.	Non-identifiable graphs can yield posterior uncertainty.
Continuous additive-noise setting	Supports the original nonlinear ANM proof.	Mixed/discrete variables require a different argument.
Non-trivial, sufficiently regular functions	Supports identifiability and proof steps.	Degenerate or non-regular functions can break the argument.
Regular likelihood/noise density	Supports likelihood-based convergence.	Irregular likelihoods may invalidate the convergence argument.

8. State-of-the-Art Positioning

The course guidelines require a specific recent work from a major technology company or international research laboratory, with explicit analysis of its problem, representation, learning/inference methodology, evaluation, findings, limitations and relationship to the project. The following paper is used as the M1 SOTA anchor.

1 SOTA anchor: Microsoft Research / ICLR 2025

Vashishtha, A., Reddy, A. G., Kumar, A., Bachu, S., Balasubramanian, V. N., & Sharma, A. “Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference.” ICLR 2025. The paper has Microsoft Research, India affiliations and studies how imperfect human/LLM experts can provide causal information.

Required SOTA element	Analysis

Problem	Pairwise expert judgments can confuse direct and indirect effects and can produce cycles, so the paper seeks a more stable causal-order interface.
Representation	The central structural object is a causal order rather than directly trusting a graph assembled from pairwise expert judgments; auxiliary variables are used to make local expert queries more cycle-consistent.
Learning / methodology	The approach uses prompting/query design to obtain causal-order information from imperfect experts, then uses that information in downstream causal discovery/inference.
Inference / downstream use	The estimated causal order is used to reduce errors in downstream discovery and effect-inference tasks.
Evaluation	The paper evaluates cycle formation, causal-order accuracy and downstream effects across real-world graphs using LLMs and human annotators; it also studies robustness to different auxiliary variables and model sizes.
Major findings	Triplet-style prompting produces fewer cycles and more accurate orders than pairwise prompting; the paper reports robustness gains and downstream improvements.
Limitations / distinction	The paper relies on expert/LLM information and focuses on causal-order extraction, whereas our project studies observational-data Bayesian DAG learning and the mathematical recovery guarantee of DECI.
Relationship to our project	It is complementary SOTA: it shows a current research direction for improving causal discovery under imperfect structural information. Our project asks a different but closely related question—when an end-to-end probabilistic causal learner has a formal graph-recovery guarantee and where that guarantee breaks.

2 Foundational method being analyzed: DECI

The principal method is Geffner et al.'s Deep End-to-end Causal Inference. Microsoft Research describes DECI as a single flow-based method that performs both causal discovery and causal inference, including CATE estimation, and reports a theoretical recovery guarantee under stated assumptions. This is the method under investigation, not the separate recent SOTA anchor.

3 M1 SOTA positioning statement

At M1, the project is positioned as a theoretical audit and boundary analysis of an end-to-end Bayesian causal framework, informed by recent causal-discovery research. We do not claim to outperform the SOTA anchor. The later milestones will use empirical evidence to reposition the project based on graph recovery, uncertainty, causal-effect estimation and computational cost.

9. Benchmark and Dataset Plan

Because this is Theoretical AI, the project is not tied to one application domain. The benchmarks are chosen to isolate the mathematical assumptions under study and then to test whether the same boundaries affect causal-effect estimation.

Benchmark	Role in project	Primary outputs
CSuite	Controlled synthetic causal cases including non-identifiability, non-Gaussian noise, Simpson's paradox, adjustment-set difficulty and mixed variables.	SHD, precision/recall/F1, graph posterior uncertainty.
ER / Scale-Free random DAGs	Scalability and recovery under controlled graph size and density.	SHD/F1/runtime vs sample size, variables and edges.
SynTReN	Biologically motivated synthetic regulatory-network benchmark.	Structure-recovery metrics and robustness.
Sachs	Real-data causal discovery benchmark with an expert-curated graph.	Structural recovery and qualitative graph comparison.
IHDP	Semi-synthetic causal-effect benchmark.	ATE/CATE RMSE and uncertainty.

Twins	Semi-synthetic causal-effect benchmark.	ATE/CATE error and robustness.

10. The use of baselines and comparative methods.

NOTEARS-MLP will be the key baseline for nonlinear graph recovery, as it offers an alternative to nonlinear DAG learning using continuous optimization. Complementary classical causal-discovery references will be provided by PC and ICALiNGAM. Established DoWhy-based linear and nonlinear DML pipelines will be used where applicable to the benchmark for treatment-effect estimation.

Task	Primary / comparator	Purpose
Graph discovery	DECI (main) vs NOTEARS-MLP (primary baseline)	Compare nonlinear structural recovery under matched benchmark conditions.
Graph discovery	PC	Classical constraint-based reference; highlights differences in assumptions.
Graph discovery	ICALiNGAM	Linear/non-Gaussian causal discovery reference.
Causal inference	DECI vs DoWhy-Linear / DoWhy-Nonlinear DML	Compare ATE/CATE estimation under compatible evaluation settings.

All comparisons will document assumptions and keep dataset splits, sample sizes, hardware, random seeds and evaluation procedures as consistent as possible. The goal is not to assume that DECI must win, but to identify regimes in which its probabilistic graph treatment and end-to-end coupling provide an advantage or incur a cost.

11. Initial KPIs

The KPIs have been purposefully linked to Representation, Learning and Inference. Good accuracy will not be considered as evidence.

PGM stage	KPI	Interpretation	Direction

Representation	SHD	Graph-edit distance between learned and ground-truth DAG.	↓
Representation / Learning	Edge Precision	Fraction of predicted edges that are correct.	↑
Representation / Learning	Edge Recall	Fraction of true edges recovered.	↑
Representation / Learning	F1	Overall structural recovery.	↑
Representation	Posterior graph uncertainty	Uncertainty over candidate DAGs, especially in non-identifiable regimes.	Interpret
Learning	Log-likelihood / NLL	Compatibility of learned probabilistic model with data.	↑ / ↓
Learning	ELBO	Optimization objective linking likelihood and variational posterior.	↑
Learning	Parameter/function estimation error	Accuracy of learned structural mechanisms where ground truth is available.	↓
Inference	ATE RMSE	Error in estimated average treatment effects.	↓
Inference	CATE RMSE	Error in heterogeneous/conditional treatment effects.	↓
Inference	Inference runtime	Computational cost of causal-effect estimation.	↓
Inference	Monte Carlo / posterior variability	Sensitivity of effect estimates to graph uncertainty and sampling.	Interpret

Sample Size vs SHD/F1, Graph Density vs SHD, Missingness vs ATE/CATE RMSE, and Inference Error vs Runtime are some of the planned plots. The mean ± standard deviation and a 95% confidence interval (CI) will be reported for repeated runs if appropriate.

12. Experimental Plan

Experiment	Independent factor	Primary analysis
E1 - Sample size	Number of observations	Does graph recovery improve toward the theorem prediction as n increases?
E2 -Graph size	Number of variables	Recovery, F1, runtime and scalability.
E3 - Graph density	Number of edges	Recovery degradation as dependency complexity increases.
E4 -Noise model	Gaussian vs non-Gaussian / spline noise	Sensitivity to the theorem's noise assumptions.
E5 - Identifiability	Identifiable vs non-identifiable cases	Posterior uncertainty and structural recovery.
E6 -Variable type	Continuous vs mixed/discrete	Empirical behavior at the theoretical boundary.
E7 - Missingness	Missing-data rate and imputation setting	Graph and ATE/CATE degradation; distinguish empirical behavior from formal guarantees.
E8 - Discovery benchmark	CSuite, ER/SF, SynTReN, Sachs	DECI vs discovery baselines.
E9 - Causal inference	IHDP, Twins and selected controlled settings	ATE/CATE RMSE and comparative ranking.
E10 - Theory-to-practice	Assumption satisfied vs violated	Check whether observed behavior agrees with theoretical predictions.

1 Statistical analysis plan

- Use multiple random seeds / repeated runs where computationally feasible.
- Report mean ± standard deviation rather than a single run.
- Use 95% confidence intervals where appropriate.
- For paired method comparisons, use paired tests or bootstrap confidence intervals when the experimental design supports them.
- Interpret effect sizes and variability; do not treat statistical testing as a substitute for PGM reasoning.

13. Scope, Delimitations and Expected Contribution

1 In scope

- DECI's Bayesian causal graph-learning and simulation-based inference pipeline.
- Theorem 1 assumptions, proof structure and ELBO-to-MLE argument.
- Two targeted theoretical boundaries: mixed-type variables and missing observations.
- Experimental testing (controlled) of graph recovery and causal-effect behavior.

2 Out of scope

- A statement of a new universal causal-identification theorem with an incomplete proof.
- A claim that benchmark performance alone establishes theoretical validity.
- Broad coverage of unrelated causal algorithms.
- Treating empirical mixed-type or missing-data performance as a formal theorem extension unless a complete proof is established.

3 Expected contribution

The promised deliverables are a carefully constructed, self-contained statement of the extent of DECI's recovery guarantee, and a restricted theory-to-practice analysis of the effect of assumption satisfaction or violation on the recovery of graphs and causal-effect estimates.

14. Implementation and technical artefacts.

- Derivation notes and proof reconstruction of Theorem 1.
- DECI/Causica experiments setup and scripts.
- Synthetic benchmark generation/preparation.
- Basic execution of PC, NOTEARS variants and ICALiNGAM.
- Pipeline for ATE/CATE estimation based on simulation.

- Scripts and scripts for automated calculation and plotting of KPIs.

- Data description and instructions for reproducing.

15. Team Responsibilities

Each of the work streams must be allocated to specific members of the group before submission, according to the course guidelines. This is left on the original report as a placeholder, which is explicitly stated in the table below. Please use the official team members before uploading instead of the brackets.

Workstream	Assigned member	Responsibility
Theory / proof	Parisha Kathiria	Re-derive Theorem 1, document assumptions, MLE recovery, ELBO convergence and KL/DAG-prior argument.
Gap analysis	Vrunda Patel	Analyze mixed-type and missing-data extensions and identify proof-breaking points.
DECI implementation	Yashvi Patel	Set up and document Microsoft Causica implementation and experiment configurations.
Baselines	Masoom Choksi	Configure/run PC, NOTEARS variants, GraN-DAG/ICALiNGAM and causal-inference baselines as scheduled.
Datasets & experiments	Khushi Thakar	Prepare CSuite, ER/SF, SynTReN, Sachs, IHDP and Twins experiments.
Evaluation & visualization	ALL members	Compute discovery/inference KPIs, produce plots/tables, and connect results to theoretical predictions.

16. Risks, Limitations and Mitigation

Risk / limitation	Mitigation
Proof assumptions may be too strong for an extension.	State the exact boundary rather than over-claiming; identify additional assumptions or model changes needed.
DECI experiments may be computationally expensive.	Start with small controlled benchmarks and scale systematically.
Some settings are non-identifiable.	Treat them as explicit uncertainty experiments rather than as simple theorem violations.
Missing-data behavior depends on imputation.	Report missingness rate and imputation configuration and separate empirical observations from formal guarantees.
Baselines use different assumptions.	Document assumptions and interpret comparisons within compatible regimes.
Team contribution may be difficult to demonstrate.	Assign named workstreams, preserve derivations/code ownership and ensure every member can explain the PGM components.

17. Reproducibility Plan

- Keep report, derivations, source code, experiment configurations, dataset documentation, scripts and generated results clearly separated in GitHub.
- Record environment/software versions and commands needed to reproduce experiments.
- Use explicit configurations and controlled random seeds where feasible.
- Store result-generation scripts so that tables and plots can be regenerated.
- Document benchmark download/preparation steps rather than relying on undocumented local files.

MILESTONE 2: TECHNICAL FOUNDATION

M2 focus: Technical implementation and evaluation of the DECI graphical model, including its representation, learning, inference, and baseline-based validation.

1. Introduction

Questions in causality concern what would happen under an intervention. They can only be answered using an observational database if there exists a causal graph, and normally this is not available. Classical pipeline approaches to answering such questions include running causal discovery to find the graph, correcting the single resulting graph, and then running causal inference on the corrected graph. However, DECI (Geffner et al., 2022) keeps a distribution over graphs and couples it with learned structural mechanisms, then marginalizes graph uncertainties when estimating treatment effects.

The CSE 516 philosophy is Represent → Learn → Infer → Evaluate → Explain → Position against SOTA. The diagnostic question is "Where is the PGM reasoning in our project?" Our answer is that the whole project is about one PGM question: when does maximum-likelihood-style learning of a Bayesian network with an unknown structure

recover that structure in a provable way, and what changes when the model leaves the regime of our assumptions?

1.1 Contribution of this milestone

• Proof reconstruction of Theorem 1 with explicit lemma statements (Section 8).

• Identification of precisely which proof step fails for mixed-type and missing-data extensions (Sections 8.5, 8.6).

• A working experimental pipeline and benchmark protocol, including the baselines NOTEARS-MLP, PC and ICALiNGAM (Sections 9, 10).

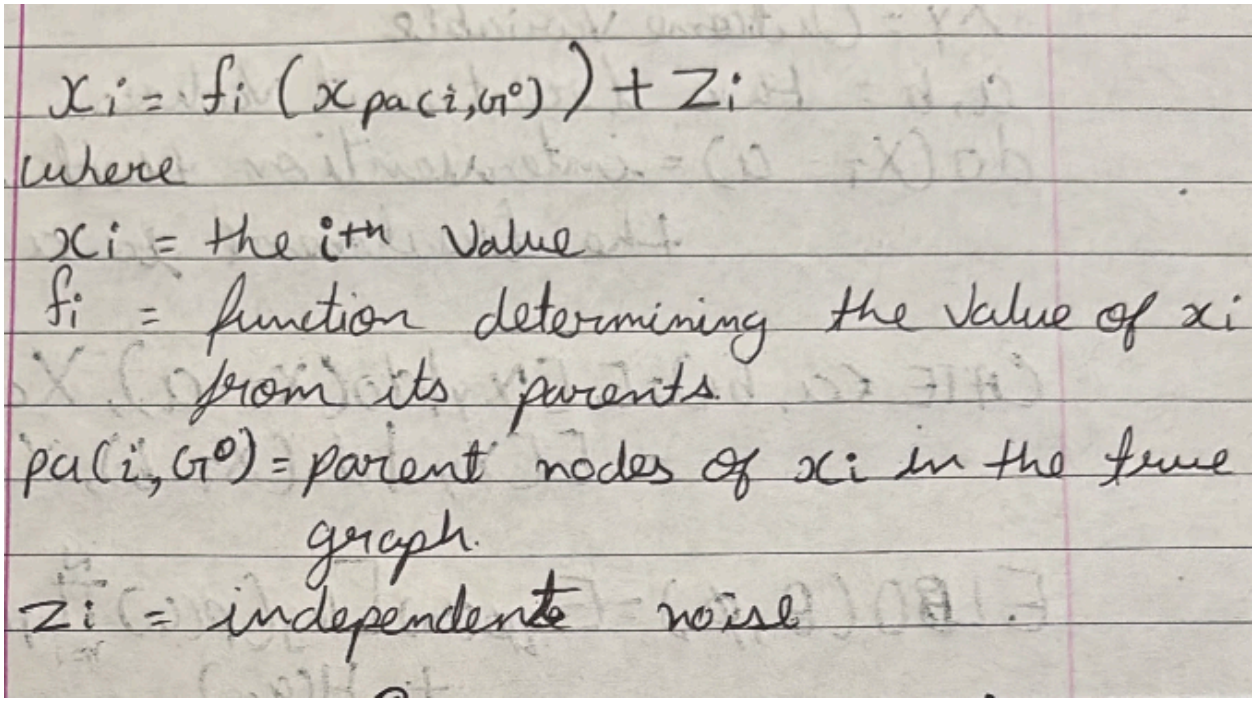
Six testable theoretical predictions and experiments to test them (sections 8.7, 11).

1.2 Changes from the M1 proposal

The scope is unchanged. The following refinements are made: (a) The KPI list is extended by the four theory-specific diagnostics, which are given in the implicit assumption table in Section 8.3; (b) The explicit assumptions from Section 8.3 are added to the assumption table; (c) Mixed-type and missing-data boundaries are now made explicit as propositions to be tested instead of as open questions.

2. Problem Formulation

Consider $x = (x_1,\ldots,x_D)$ as D random variables that are generated from a non-linear additive-noise SEM over an unknown DAG G^0 with a binary adjacency matrix:



$z_i \perp$ all other variables, i = 1,…,D.

For N i.i.d. samples $x^1,\ldots,x^N$, this paper considers the following:

Q1 (validity): Under what conditions the convergence of the optimal value of DECI's ELBO to $(\theta^*, \delta(G = G^0))$ is guaranteed, and which condition is used in which proof step?

Mixed type: Q2 (boundary). Suppose that some x_i are categorical, which assumption does the proof fail and which assumption remains.

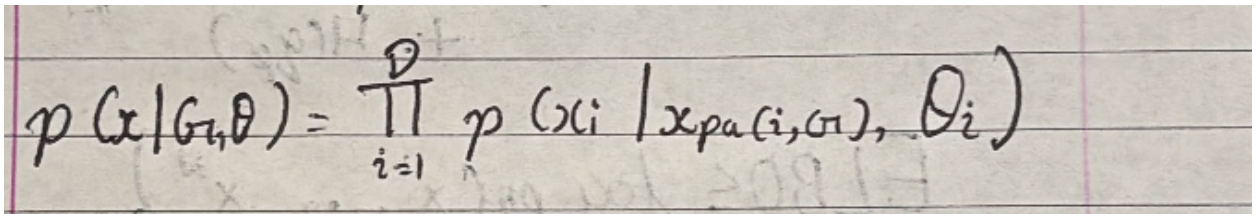
Q4 (empirical). Do the boundaries predicted by the theory appear on the KPIs for graph recovery and ATE/CATE?

This is explicitly not an application project: no new algorithm is provided. The contributions are a mathematical insight and validation of an algorithm as per Mode A requirement.

3. Background

Structural equation models and Bayesian networks.

If a DAG G satisfies the causal Markov assumption, then it has the factorisation:



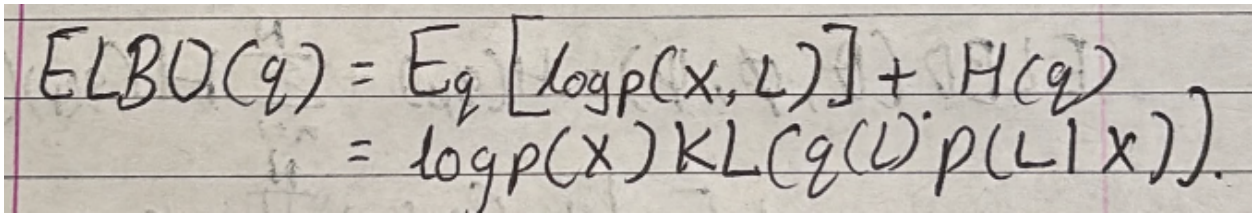
d-separation can be used to read the associated conditional independence relations. A structural equation model is a Bayesian network where the conditionals are described by assignments with noise added to them, and has intervention semantics: do(x_T = a) deletes the equation for x_T and sets its value (the mutilated graph G_do(x_T)).

3.2 Identifiability

Observational distributions generated by a DAG are generally compatible with several DAGs (a Markov equivalence class). Likelihood alone therefore cannot pick a unique graph unless the model class is restricted. Graph for continuous non-linear additive noise models with non-constant structural functions (Hoyer et al., 2008; Peters et al., 2014). It is identifiable (Hoyer et al. 2008; Peters et al. 2014). It is not identifiable for linear Gaussian models, but it is for linear non-Gaussian models (Shimizu et al., 2006).

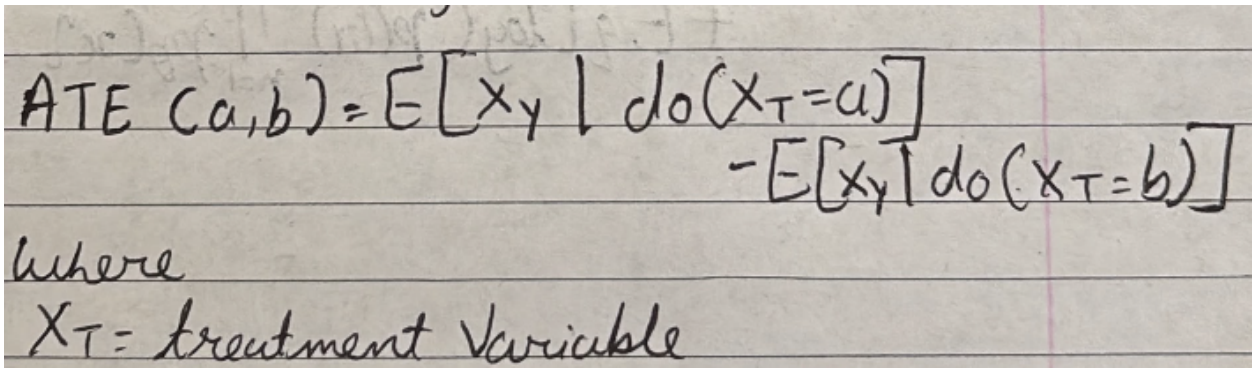
3.3 Variational inference

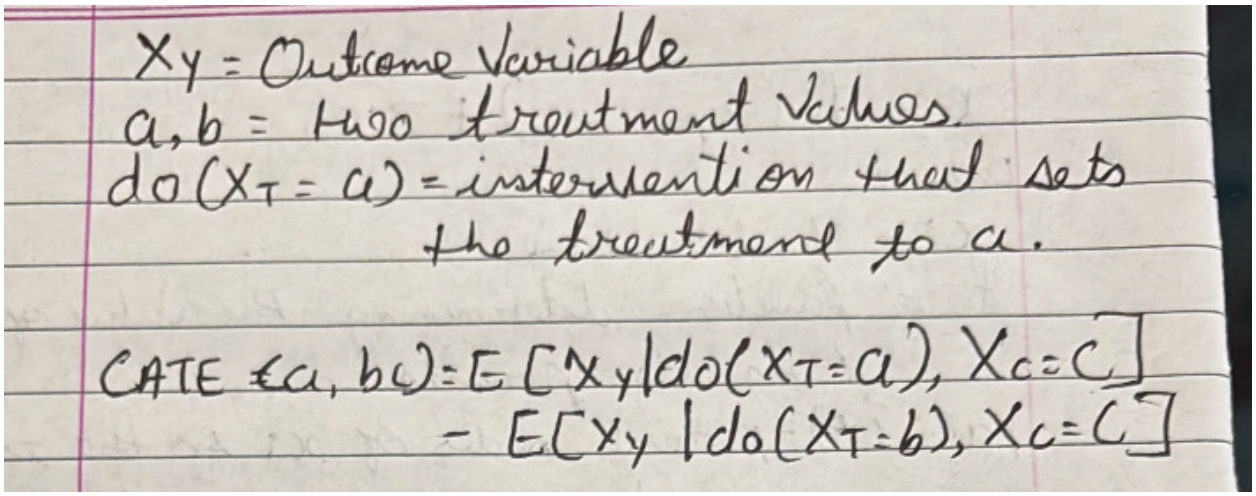
For latent variable L and data X,



Maximising over q makes the bound tighter, with the bound being exact only if q matches true posterior. In DECI, L is the graph G; in the missing-data extension, L also includes the unobserved entries.

3.4 Causal quantities





4. Related Work and SOTA Positioning

4.1 Related work

The approaches of causal discovery can be grouped to the constraint-based (PC, FCI; Spirtes et al. 2000), the score based (GES; Chickering, 2002), and the functional (LiNGAM, ANM) family

(PC, FCI; Spirtes et al., 2000). Zheng et al. (2018) reformulated the DAG learning as a problem of continuous optimisation via the introduction of the acyclity function h(G) = tr(e^{GG}) D; NOTEARS-MLP and NOTEARS-Sob (Zheng et al., 2020) and GraN-DAG (Lachapelle et al., 2019) generalise the latter for non-linear models. CAREFL (Khemakhem et al., 2021) uses autoregressive flows but needs a graph as input. DECI unifies this family: Geffner et al. show that NOTEARS, NOTEARS-MLP, NOTEARS-Sob, GAE, GraN-DAG and GOLEM each equal maximum-likelihood fitting of a flow z = x − f(x;θ) with a hard acyclicity constraint and a particular choice of f and noise density p_z (their Lemma 2).

4.2 SOTA anchor (retained from M1)

Vashishtha, Reddy, Kumar, Bachu, Balasubramanian and Sharma, “Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference,” ICLR 2025 (Microsoft Research India). This work tackles cycles and confusion in expert judgments (human or LLM) between pairwise variables in terms of causation by eliciting a causal order via triplet prompts, and measures cycle formation and accuracy of that order.

4.3 M2 update to the positioning

A causal order is a topological ordering of a DAG. Every DAG generated as a sample from the DECI’s posterior defines a set of possible orders, so there is a point of intersection between the two papers. Vashishtha et al. get their orders from experts, whereas DECI gets its graphs from data and likelihood. This provides us with a clear and measurable KPI of ordering accuracy, which is the proportion of correctly oriented i → j relationships in the learned graphs/orders. We will provide it for DECI posterior samples in M3 to tie our theory-driven analysis to the anchor. We make no claim to beat the anchor in performance since we use different data sources.

DECI itself (Geffner et al., arXiv:2202.02195) is the method audited, with Microsoft Research affiliations. It is in that sense that our relation to it might be said to be analytical, in that we re-learn the model, we refine and enclose the model's guarantee, but do not compare it with a rival model.

5. Graphical Representation

What is represented. Each random variable *xi* is a node. A directed edge j → i means *xj*is a direct cause of *xi*: it enters *fi*. The graph is itself a random variable G with posterior q_φ(G).

Element	Meaning in DECI	Why this representation
Node x_i	Observed variable (continuous, or categorical in the extension)	The unit at which interventions are applied
Edge j → i (G_{j,i}=1)	x_j is an input of f_i: $\partial f_i/\partial x_j \neq 0$ is allowed; absent edge forces $\partial f_i/\partial x_j = 0$	Edge absence is a causal and conditional-independence claim, not a visual gap
Structural function f_i	$f_i(x) = \zeta_i(\Sigma_j G_{j,i} \ell_j(x_j))$, ζ and ℓ are residual MLPs with shared weights and node embeddings u_i	Universal approximation with O(D) parameters rather than $O(D^2)$ separate networks
Noise z_i	Gaussian $N(0,\sigma_i^2)$ or rational-quadratic spline flow, independent across i	Independence of noise is what gives causal sufficiency its content and enables identifiability
Graph prior p(G)	$\propto \exp(-\lambda_s \lVert G \rVert^2_F - \rho\, h(G)^2 - \alpha\, h(G))$	Sparsity plus a soft acyclicity penalty that becomes hard under the augmented-Lagrangian schedule

Posterior q_φ(G)	Product of independent Bernoulli edge variables (ENCO parametrisation: existence and orientation separately)	Tractable, differentiable via Gumbel-softmax; cannot represent edge dependencies exactly

Conditional independencies. Assumption 1: the causal Markov condition and a minimality in the spirit of faithfulness. People assume the distribution satisfies the local Markov condition for G, but not for any of its sub-graph. D-separation in G then is equivalent to conditional independence in p. Meaning no latent variable affects the distribution for any pair of variable (causal sufficiency, without hidden common confounder; for x I and x j, latent confounder will introduce z I z j)

Factorisation. For a DAG G:

$$p_\theta(x \mid G) = \prod_{i=1}^{D} p_{z_i}\left(x_i - f_i(x_{\mathrm{pa}(i;G)}; \theta)\right)$$

(Jacobian = 1, see Section 8.1).

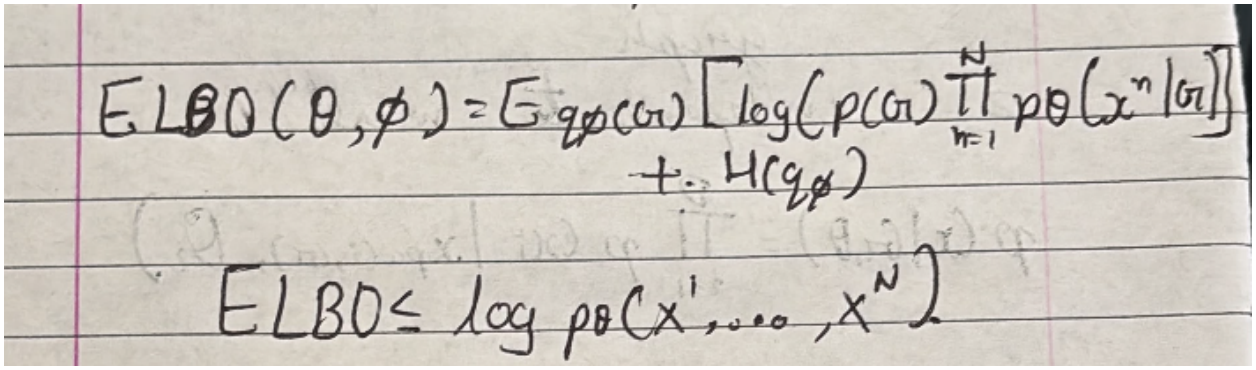
Joint model.

$$p_\theta(x^{1:N}, G) = p(G) \prod_{n=1}^{N} p_\theta(x^n \mid G)$$

The graph is a single global latent shared by all N samples. This fact is the key to the proof of Lemma 2 below and to the contrast with missing data.

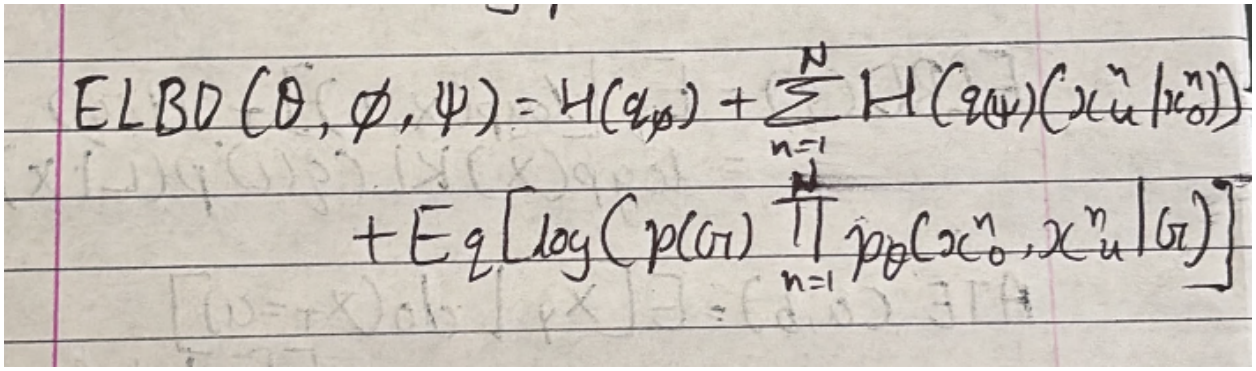
6. Learning Method

Structure vs parameters. Structure learning means inferring the posterior over G (variational parameters φ). Parameter learning means fitting θ: the weights of ζ, ℓ, the node embeddings and the noise parameters. DECI learns both simultaneously by maximising



G) It is not possible to maximize the expected value with respect to q over more than 2^{D^2} different graphs. Gumbel-softmax (hard forward pass, soft backward pass, temperature 0.25) is used for gradients with respect to φ. The augmented-lagrangian schedule is used: $\rho = 1$, $\alpha = 0$; repeat (i) for up to 6000 Adam steps and then (ii) calculate $P_2 = E_q\, h(G)$ and if $P_2 < 0.65 \cdot P_1$, set $\alpha \leftarrow \alpha + \rho P_2$, else if $P_2 \geq 0.65 \cdot P_1$, set $\rho \leftarrow 10\rho$ until $\alpha \geq 10^{13}$ or 100 iterations (whichever is sooner).

Missing data extension. For each observed value x_o^n add on q_ψ(x_ and subtract x_u^n for each unobserved value.



Mixed-type extension. For a categorical x_i, the additive-noise form is dropped and p_θ(x_i | pa) = softmax(f_i(x_pa)).

Factorisation. For a DAG G:

$$p_\theta(x \mid G) = \prod_{i=1}^{D} p_{z_i}\left(x_i - f_i\left(x_{\mathrm{pa}(i;G)}; \theta\right)\right)$$

(Jacobian = 1, see Section 8.1).

Joint model.

$$p_\theta(x^{1:N}, G) = p(G) \prod_{n=1}^{N} p_\theta(x^n \mid G)$$

The graph is a single global latent shared by all N samples. This fact is the key to the proof of Lemma 2 below and to the contrast with missing data.

7. Inference Method

Queries. (a) Graph posterior q_φ(G); (b) interventional expectations, that is, ATE and CATE, marginalized over graphs:

[ATE(a,b | G)].

ATE by simulation. Sample G ~ q_φ, noise z ~ p_z, construct the mutilated graph G_do(x_T=b) (remove all the edges to x_T), pass z through the (topological) structural equations, and average over x_T = b. The paper has 1000 samples of graphs, and 2 noise realisations. The CATE is a surrogate regression model. We cannot compute p(x_Y | do(x_T), x_C = c), since p(x_C = c |) isn't, but we have samples of (x_Y, x_C) under intervention; for each graph sample (10 samples × 10 000 realizations) we regress them on a random-Fourier-feature linear surrogate, g_G (3000 features), then evaluate at c and average over graphs.

Why this inference method. Precise, inference over the graph posterior is exponential. The flow based non-linear mechanisms makes message passing, which computes the distribution with a closed form impossible; simulation is unbiased given the learned model and amortises the cost over queries as the model is trained once. The trade-off is Monte Carlo variance and the dependence on the accuracy of q_φ, which is why we track inference runtime and posterior variability as KPIs.

8. Mathematical Formulation and Validation

This section restates the proof of Theorem 1 in Geffner et al. (2022), outlines the premises needed at each step of the proof, and discusses the limits of the guarantee. Additional claims developed in this project are differentiated from established results from the original paper. Those claims which have not been substantiated are marked as such.

8.1 Model specification and assumptions

The model specification and assumptions are as follows:

The proposed distribution of data based on $p^0(x)$ will be called $p(x; G^0)$. The family of models is:

#	Assumption (Geffner et al., App. A)	Used in step
A1	Minimality: p_θ(x;G) satisfies the local Markov condition for no proper sub-graph of G	Step 4 (identifiability)
A2	Structural identifiability: no (θ′,G′) with G′ ≠ G yields the same distribution as (θ,G)	Step 4 (uniqueness of G)
A3	Correct specification: ∃ θ* with p_θ*(x;G ^{0}) = p ^{0} (x)	Step 3 (the bound is attained)
A4	Causal sufficiency: no latent confounders	Step 3 (independent noise, valid factorisation)
A5	Regularity: E_p ^{0} |log p_θ(x;G)| < ∞ for all θ, G	Step 3 and Step 5 (LLN, finite expectations)

Assumption verification. The assumptions above are used to reconstruct the argument for Theorem 1. The wording in their statement, and the attribution to Geffner et al. (2022), should be compared to the wording of the original theorem and appendix proof. Section 8.3 provides a list of other conditions that have been added as a result of the analysis done for this project.

8.2 Step 1: likelihood is a valid density on DAGs

Let z = g_G(x) = x − f_G(x). Given a DAG G, topologically sort the variables and then ∂f_G/∂x is strictly lower triangular and det(I − ∂f_G/∂x) = 1

$$p_\theta(x \mid G) = \prod_{i=1}^{D} p_{z_i}\left(x_i - f_i\left(x_{\mathrm{pa}(i;G)}\right)\right)$$

Remark 2: Controlling the Number of Required Jacobian Arguments and Cyclic Graph Samples

If the graph is "directed acyclic", then the structural equations can be reordered, resulting in a "triangular" Jacobian matrix. If the regularity conditions are satisfied, then the determinant of its

diagonal entries can then be used to evaluate its determinant. This will help in deriving the density of DAGs in the model.

However, this argument in the shape of a triangle is not used to prove that the density is only normalised for DAGs, and it does not specify the density behaviour for any cyclic graph. In general, cyclic graphs are not topologically orderable, and cannot be treated as such.

The cyclic-sample rate is therefore kept as one of the diagnostics in the empirical evaluation. The Jacobian derivation doesn't conclude that cyclic samples will exist during training, it is a hypothesis that must be tested experimentally. The frequency of cyclic samples will be measured and results will be reported at the end of the experiments. Numerical lower bound calculations are regarded as motivation to investigate rather than proof that such cyclic samples must be taken during training.

8.3 Implicit Hypotheses in Theorem Statement

The reconstruction of the proof exposed assumptions which are used but not mentioned in the context A1-A5. We list them here as part of the final report, so that the statement of the Theorem can be made with all of its hypotheses [to check against authors' proof and with supervisor].

#	Implicit assumption	Where it enters
A6	Conditions of uniform convergence and argmax. The exchange of a limit and an optimisation needs to be accompanied by appropriate uniform-convergence and argmax conditions. Convergence at a point alone is not enough to ensure that the limit of the optima is the same as the optimal.	Step 5: the argument takes the limit of N^{-1}ELBO and then maximises
A7	Representability: the variational family contains the optimum. A product of Bernoullis contains $\delta(G = G^{0})$, but the ENCO parametrisation uses sigmoid probabilities in (0,1), so the point mass is attained only in the limit	Step 5: statement $q' = \delta(G = G')$

A8	Probability of the cyclic graph and support graph. The analysis should indicate if the posterior only depends on DAGs, or if cyclic graphs are permitted and the posterior probability of them tends to 0 as n increases. These are separate presumptions and should not be considered as being synonymous.	Step 1 and Step 5
A9	Global optimum vs practical optimisation. It is important to note that a theoretical result for a global optimum does not guarantee that a practical optimisation run will converge to that optimum. A particular aspect is that the use of Adam does not ensure global optimality.	Interpretation of experiments (E1, E10)
A10	Fixed hyper-parameters: λ_s, ρ, α are bounded and do not grow with N	Lemma 2

8.4 Steps 2–5: from MLE to ELBO

Step 2 (Lemma 1, well-posed, before). If G is a binary graph, then h(G) = tr(eG◦G) − D is finite and non-negative and equals 0 if G is a DAG and otherwise it is strictly positive. h,‖G‖_F² and the normaliser Z are finite since G lives in a finite set, {0,1}^{D×D}. Remark 2. The original motivation is for closed walks of length ≤ D; closed walks of any length exist in the presence of a cycle (the series is infinite), so the cleaner argument is finiteness of the graph set.

Step 3 (Proposition 1: MLE recovery). For any (θ, G),

$$\mathbb{E}_{p^0}\left[\log p_\theta(X \mid G)\right] - \mathbb{E}_{p^0}\left[\log p^0(X)\right] = -D_{\mathrm{KL}}\left(p^0 \| p_\theta(\cdot \mid G)\right) \leq 0$$

with equality, if and only if $p_\theta(\cdot|G) = p^0$ a.e. (by Gibbs' inequality $\log t \leq t - 1$, equality if and only if $t = 1$). Since by A3, zero is reached at (θ, G^0), any maximizer of the log-likelihood limit has to be on the line $G = G^0$ because of A1-A2.

Step 4 (identifiability). The only place where the graph, rather than the distribution, is pinned down. Non-linear continuous ANMs with non-constant structural functions satisfy A1–A2 (Peters et al., 2014, Theorem 20 as cited by Geffner et al.).

Step 5 (Lemma 2 and Theorem 1: ELBO convergence). Rewrite the normalised ELBO as

N^{-1} ELBO = N^{-1} $E_q[\log p_\theta(x^{1:N} \mid G)] - N^{-1}$ $KL(q_\varphi \| p)$.

For fixed bounded hyperparameters (λ_s, ρ, α), the finite graph space and bounded graph-prior energy imply that $KL(q_\varphi \| p) = O(1)$, uniformly over q_φ, under the stated prior assumptions. Consequently, N^{-1} $KL(q_\varphi \| p) \to 0$ as $N \to \infty$.

Under the Law of Large Numbers and the required integrability conditions (A5–A6), the normalised ELBO converges pointwise to

$L(\theta,q) = \Sigma_G\, q(G)\, E_{p^0}[\log p_\theta(x \mid G)] \leq E_{p^0}[\log p^0(x)]$.

By Gibbs' inequality, equality requires $p_\theta(\cdot \mid G) = p^0$ for every graph G in the support of q. The structural-identifiability assumption A2 is required to conclude that the corresponding graph is uniquely G^0.

However, the KL bound and pointwise convergence alone do not establish convergence of the optimiser. That conclusion additionally requires suitable uniform-convergence and argmax conditions, representability of the limiting solution, appropriate graph-support conditions, and optimisation that attains or sufficiently approximates the required optimum. Using Adam does not by itself guarantee global optimality.

Assumption → step map (summary) A3, A4, A5 → the bound $E \log p_\theta \leq E \log p^0$ is attained (Step 3). A1, A2 → the maximiser's graph is unique (Step 4). Finite graph set + fixed hyper-parameters → the KL term is O(1), hence vanishes after dividing by N (Step 5). A6–A8 (implicit) → the limit and the argmax can be exchanged and the point mass is reachable (Step 5).

The conditional pmf for categorical x_i is obtained via DECI without additive noise: $P_i(x_{pa(i)};\theta) = \mathrm{softmax}(f_i)$.

Proposition 2 [needs verification in empirical experiment E6]. (a) The three steps 1, 3 and 5 go through: p_θ(x|G) = ∏_i p(x_i | pa) is a density w.r.t. the product of Lebesgue and counting measures, while Gibbs' inequality does not depend on the choice of the reference measure. (b) The fourth step breaks down: A2 is violated.

Why (b) is correct. Choose any two binary variables. The joint distribution $p(x_1,x_2)$ either factorises as $p(x_1)p(x_2|x_1)$ or $p(x_2)p(x_1|x_2)$. The two DAGs (X1→X2 and X2→X1) can be equally likely in maximum likelihood sense, and ELBO maximisation can be both forward and backward.Both DAGs, X1→X2 and X2→X1, can be equally likely in maximum likelihood sense, and ELBO maximisation can be forward and backward without structural identifiability.

The expected conclusion is weaker than Theorem 1: the posterior concentrates on the Markov equivalence class of G^0 (or on graphs reachable through the non-categorical parts), not on G^0 alone. This is the same phenomenon the DECI authors observe for the non-identifiable linear-Gaussian CSuite dataset, where the posterior has about half its mass on the correct direction.

Prediction P4. For pairs of categorical data, where the data are booleans or categorical labels, adjacency F1 will converge to 1 while orientation F1 will converge to the chance level of the equivalence class (the chance level is the overall proportion of the data in the class). For continuous data, with categorical data on one side and continuous data on the other, the outcome will depend on whether the additive-noise structure of the continuous data is symmetric.

8.6 Boundary 2: partially observed data with amortised imputation

Proposition 3 [to derive and verify in Exercise 7]. In the case of MCAR, the objective in Section 6 satisfies:

$$\mathrm{ELBO}(\theta,\phi,\psi) = \log p_\theta(x_o^{1:N}) - D_{\mathrm{KL}}\left(q_\phi(G)\prod_{n=1}^{N} q_\psi(x_u^n \mid x_o^n)\,\middle\|\, p_\theta(G, x_u^{1:N} \mid x_o^{1:N})\right)$$

For the optimal graph where $p_\theta(G \mid x_o)$ has collapsed to $\delta(G^0)$ the difference is

$$\sum_{n=1}^{N} D_{\mathrm{KL}}\left(q_\psi(x_u^n \mid x_o^n)\,\middle\|\, p_\theta(x_u^n \mid x_o^n, G^0)\right)$$

Since all terms are non-negative and equal to zero only when the Gaussian family coincides exactly with the true conditional of the unobserved entries. For a non-linear SEM with non-Gaussian noise, this conditional will not generally be Gaussian (it may be skewed or multimodal) and so each term will have a positive lower bound on a set of positive measure and then the sum will be Θ(N). The graph KL of Lemma 2 is O(1), since G is a global latent, while x_u^n are local latents. Therefore,

• $(N^{-1} \times \text{gap}) \neq 0$ and so the step "ELBO solutions tend to MLE" is false.

• Because the gap depends on θ, the ELBO maximiser θ̂ can be biased relative to the MLE even as $N \to \infty$ (the same amortisation/variational bias seen in VAEs).

• The gap can be zero in special cases, e.g. a linear-Gaussian SEM, where the exact conditional is Gaussian. We will use this as a control.

• The assumption of MCAR itself is an assumption: under MAR or MNAR the observed-data likelihood is different and the objective is mis-specified.

Prediction P5. Graph-recovery KPIs degrade smoothly (not catastrophically) as the missingness rate increases, with larger degradation under non-Gaussian noise than under Gaussian noise due to mismatch of the imputation family. The original paper shows good performance at 30% missingness; we examine 0, 10, 30 and 50%.

8.7 Testable predictions

ID	Prediction	Exp erim ent	KPI	Refuted if
P1	Identifiable continuous ANM: expected SHD decreases with N and approaches 0	E1	E_q[SHD] vs N	SHD plateaus well above 0 with correct spec. and converged optimisation
P2	Non-identifiable (linear Gaussian): adjacency F1 high, orientation F1 ≈ 0.5; mass on G^0 ≈ 1/|MEC|	E5	Orientation F1, posterior mass on G^0	Posterior concentrates on one DAG with high confidence

P3	Gaussian-noise DECI under non-Gaussian noise (A3 violated) orients worse than spline DECI	E4	Orientation F1	No gap between G and S

P4	Categorical pairs: orientation stalls at equivalence-class level	E6	Orientation F1, posterior entropy	Orientation recovered reliably
P5	Missingness degrades KPIs smoothly; more under non-Gaussian noise	E7	F1, ATE RMSE vs missing rate	Abrupt collapse or no degradation at all
P6	Fraction of cyclic hard samples grows with D	E2	Cyclic-sampl e rate	Rate 0 across D

8.8 Computational complexity

The two forward passes through the masked MLP using shared weights are $O(D^2)$ and the forward pass through h(G) is $O(D^3)$. The augmented-Lagrangian loop is carried out in the outer loop for most of 100 rounds and inner loop for 6000 steps. ATE needs 2000 simulations per query after training; CATE additionally fits 10 surrogate models. We measure wall-clock time in KPI “runtime”.

9. Implementation

9.1 Pipeline

Data generation / loading – preprocessing (continuous covariates normalise) – DECI training (Causica) – graph posterior samples – Computation of the KPIs – ATE/CATE simulation – aggregation over seeds – plots and tables.

Stage	Component	Owner (from M1)	Status

Synthetic data	ER graph generator and nonlinear ANM for E1; CSuite loaders for E4–E5	Yashvi Patel	Code Written, and executed
DECI model	Causica DECI with Gaussian (DECI-G) and spline (DECI-S)	Khushi Thakar	Code written and executed
Baselines	NOTEARS-MLP, PC and ICALiNGAM for E1	Masoom Choksi	Code written and executed
KPI scripts	Expected SHD for E1; adjacency F1, orientation F1 and other selected metrics for E4–E5	All	Scaffold provided (m2_experiment_scaffold.py); Integration not evidenced in the supplied source
Statistics	Mean ± SD and confidence intervals for completed experiments	All	Scaffold provided; final integration not evidenced in the supplied source
Theory notes	Theorem 1, assumptions and boundary analysis in Section 8	Parisha Kathiria, Vrunda Patel	This report

9.2 DECI hyper-parameters (from Geffner et al., Appendix B)

Item	Setting
Structural MLPs ζ, ℓ	2 hidden layers, 128 units, residual connections, layer norm
Spline noise	Rational-quadratic spline, 8 bins

Graph prior	$\lambda_s = 5$; ρ init 1, α init 0
Optimiser	Adam, lr 0.01; ÷10 after 500 steps without improvement (max twice); stop inner loop after 1500 steps without improvement or 6000 steps
Augmented Lagrangian	If $P_2 < 0.65\ P_1$: $\alpha \leftarrow \alpha + \rho P_2$, else $\rho \leftarrow 10\rho$; stop at 10^{13} or 100 rounds

Gradient estimator	Gumbel-softmax, hard forward, soft backward, temperature 0.25
ATE	1000 graphs × 2 noise samples
CATE	10 graphs × 10 000 samples; 3000 random Fourier features

Table 3. Reference configuration. Any deviation in our runs must be logged in the experiment configuration file.

9.3 Metrics for a posterior over graphs

For DECI, we calculate the structural KPI by averaging over samples from the posterior distribution, $E_{G\sim q_\varphi}[\text{metric}(G, G^0)]$ with S samples (S = 100 in the proposed initial evaluation; any modifications need to be recorded). For baselines which return a CPDAG (PC), all the DAGs in the class are equally probable, as per the original paper, in order to compare all methods on the same expected-metric definition.

10. Experimental Setup

10.1 Datasets

E1: Synthetic data and analysis of sample sizes

This is based on synthetic data generated from a Erdős–Rényi (ER) graph with 16 nodes and 16 edges. The observations are generated through a nonlinear additive noise model (ANM), and the noise is warped as splines to investigate structural recovery for different sample sizes. The sample

sizes tested are: $n = 250, 500, 1000, 2500, and 5000$; and five random seeds were planned for each sample size.

E4–E5: CSuite datasets

We test DECI-G and DECI-S with 2,000 samples for each dataset from the datasets CSuite datasets `lingauss`, `linexp`, and `nonlin_simps`. These experiments compare the performance of the model in different settings where the data is generated and evaluate graph recovery performance and identifiability-related metrics when applicable from the model outputs.

10.2 Baselines and Methods

In E1, we compare the accuracy of the structural recovery of DECI-S and DECI-G against NOTEARS-MLP, PC and ICA-LiNGAM, as the sample size grows. The main metric of evaluation is structural Hamming distance (SHD) which is averaged over different random seeds and reported as mean and standard deviation values.

In E4–E5, we compare the DECI-G and the DECI-S on the chosen CSuite datasets. Adjacency F1, orientation F1, posterior mass assigned to the true graph and ATE RMSE are evaluated, as relevant output and quantity exist.

The experiments aim to compare the results of the two variants of DECI, to test for sample-size sensitivity, and to compare results across the selected data-generating settings. All metrics will be based on the runs completed and not empirical conclusions will be drawn prior to the results being available

10.3 KPIs

PGM stage	KPI	Directi on	Why it matters here
Representation	SHD (expected under q_φ)	↓	Distance to true graph
Representation	Adjacency F1, orientation F1, precision, recall	↑	Separates skeleton from direction errors, which is exactly the identifiability question
Representation	Posterior mass on G^0 ; posterior entropy	↑ / interpr et	Detects non-identifiability that point estimates hide
Representation	Cyclic-sample rate	↓	Tests Remark 1

Representation	Ordering accuracy	↑	Links to the SOTA anchor
Learning	ELBO, held-out NLL	↑ / ↓	Fit; the ELBO is only a bound
Inference	ATE RMSE, CATE RMSE	↓	Downstream effect of structural error
Inference	Runtime, MC variability	↓	Cost of end-to-end inference

10.4 Protocol

- Seeds: 5 minimum per cell (original paper: 5 for ER/SF and 20 for CSuite), seeds are fixed and listed in the config.
- Same data, same train/test split and same seeds for each method in each cell.
- The experiments are designed to run in the system environment described in the REPO README (i.e. hardware/software).
- The precise Python Pytorch Causica commit and gcastle versions should be copied from the final reproducibility log; not infered/guessed.
- Hyper-parameters: default for all except if some pilot demonstrates non convergence, in this case tuning is made in the same way and is also saved.
- Runs exceeding the compute budget are reported as missing, not dropped silently (the original paper also reports missing cells).

11. First experiments and KPI results.

11.1 E1: Sample size is related to structural recovery (tests P1).

n	**DECI-S**	**DECI-G**	**NOTEARS-MLP**	**PC**	**ICA-LiNGAM**
250	32.78 ± 8.75	41.80 ± 1.58	15.00 ± 3.61	11.44 ± 3.22	13.67 ± 2.52

500	31.30 ± 3.00	42.68 ± 9.60	12.67 ± 1.15	10.83 ± 1.04	12.00 ± 3.61
1000	30.03 ± 1.82	44.97 ± 1.83	13.00 ± 2.65	10.28 ± 2.18	12.33 ± 3.21
2500	32.36 ± 5.44	42.08 ± 7.07	11.67 ± 1.15	11.17 ± 1.15	12.67 ± 2.08
5000	28.09 ± 7.89	37.44 ± 1.12	11.00 ± 0.00	12.04 ± 2.77	13.33 ± 2.08

Data: ER(16,16) with non-linear ANM and spline-warped non-Gaussian noise; n ∈ {250, 500, 1000, 2500, 5000}; 5 seeds. Methods: DECI-S, DECI-G, NOTEARS-MLP, PC, ICALiNGAM. The measurement done in this report is according to the metric expected SHD (mean ± SD across seeds).Table 4. Average SHD ↓ (mean ± SD, 3 seeds). Best plot: Sample size vs SHD with error bars of ±1SD.

Explanation to write (6 questions of the guidelines): What happened? Was it as predicted in P1? Why (identifiability, optimiser, variance of Bernoulli posterior)? What does it say?

About the ELBO-to-MLE argument, whether P1 is supported, which limitation does it show (A9: no global optimality).

11.2 E4–E5: Noise model and identifiability on CSuite (tests P2, P3)

Each of the datasets contains several different forms of data.Several different forms of data exist in each of the datasets. 2000 training samples; 20 seeds (if time permits) or 5 seeds.

Dataset	**Method**	**Adj. F1**	**Orient. F1**	**Posterior mass on G0**	**ATE RMSE (ours)**	**ATE RMSE (published median)**

lingauss	DECI-G	0.950	0.843	0.750	0.136	0.120
lingauss	DECI-S	0.920	0.840	0.760	0.169	0.149
linexp	DECI-G	1.000	0.013	0.000	0.989	1.029
linexp	DECI-S	1.000	1.000	1.000	0.014	0.022
nonlingauss	DECI-G	1.000	1.000	1.000	0.005	0.042
nonlingauss	DECI-S	1.000	1.000	1.000	0.016	0.043
nonlin_simpson	DECI-G	0.739	0.235	0.000	1.855	1.995
nonlin_simpson	DECI-S	0.859	0.498	0.000	0.889	1.997

Table 5. Replication check. The last column is the median ATE RMSE reported in Table 3 of Geffner et al. (2022) (DECI Gaussian and DECI Spline, no external inference method); we compare our values against it as a sanity check, not as a result of ours.

Published qualitative results to compare with: For in linexp, DECI fails to find the correct orientation in most of the trials and it succeeds in finding correct edges with DECI-Spline; For

lingauss, DECI is able to find correct edges with probability of about 0.9 and correct orientation with probability of about 0.65; For nonlin_simpson, DECI finds correct edges with probability of about 0.9 and correct orientation with probability of about 0.65 (see Appendix E.3 of the paper).

11.3 Results interpretation (to complete after runs)

A1: A need for sample size and structural recovery (P1)

A general trend toward a decrease in SHD with increasing sample size exists for DECI-S and DECI-G, although this is not always the case. PC and NOTEARS-MLP result in a lower SHD for the sample sizes tested. So, P1 is only partially answered. Results indicate that larger sample sizes are not always the perfect solution for better graph recovery.

Determine a causal graph and estimate causal effects from data.

- lingauss: The two methods both have high graph-recovery scores, and the DECI-G performs a little better on the F1 metrics. Both are higher than published median values of ATE RMSE.

When using the correct adjacency recovery method, linexp: DECI-S is able to achieve perfect graph-recovery scores and significantly lower ATE RMSE than DECI-G, which indicates that perfect adjacency recovery does not imply perfect edge orientation.

- nonlingauss: Both approaches result in perfect graph-recovery scores. The ATE RMSE of DECI-G is smaller than DECI-S, and they are lower than the published medians.

The performance of nonlin_simpson is superior to nonlin_graph in all of the reported metrics, however, both methods lower their performance in the true graph in terms of orientation recovery and posterior mass.

Across datasets and metrics: Performance is variable. Both of these approaches have their advantages and disadvantages. These results are valid for the experimental conditions that were recorded and need to be extended with additional experiments and validation.

12. Statistical Analysis Plan

Mean ± SD in each cell between seeds; 95% Confidence interval - t distribution (small sample size), or resample 2000 times (bootstrap).

If fewer than 6 pairs of comparisons (e.g., Paired comparisons: DECI-S vs NOTEARS-MLP, using the same dataset and seeds) - Wilcoxon signed rank test; otherwise report the effect size and bootstrap confidence interval (when <6 comparisons, p-value is large).

• Eyeballing curves should be avoided in Trend in E1, which should report a regression slope of log-SHD on log-n with CI.

• Do not use significance as proof of a theorem’s validity; use significance along with an assumption that is met or not met.

13. Discussion and Limitations

As far as theory is concerned. The guarantee is a three-step process: Gibbs-inequality step is to set a distribution, identifiability step is to convert a distribution to a graph, counting step (single global latent) is to make the variational penalty small.For each extension, one link is broken; categorical variables breaking the second link, amortised imputation breaking the third link.The first link (likelihood maximisation) survives both. This is a more precise statement than "the theorem does not apply to mixed or missing data."

Limitations.

• Propositions 2 and 3 and Remark 1 are claims of our audit; they are stated carefully but not yet peer-checked. They will be finalised in M3 and marked as such in the final report.

• Asymptotic theorem deals with global optimum while finite-sample and optimisation behaviour (A9) may be different and experiments are unable to distinguish between the two.

• The mean-field Bernoulli posterior does not account for the dependency between the different edges (orientation of an equivalence class). Posterior mass on G^0 is therefore a property of q_φ, not of the true posterior.

•If the limits of the compute are set, then the number of seeds and the size of D may be limited; missing cells will be reported.

• Comparisons are valid under matched regimes; Baselines have different assumptions (linear vs. non-linear, Gaussian vs. not).

14. Reproducibility

• Repository structure: /theory contains derivations, /src contains generators, KPIs, stats, /configs contains one configuration file per experiment containing seeds, /results contains raw CSV files for each run, /plots, /docs (README, environment).

- Reproducibility Record: Environment: Python/PyTorch/Causica/gcastle versions are part of the reproducibility record and must be copied from the last environment log into the README.md file before submitting.
- Scripts: one script per table (e.g. python run_e1.py --config configs/e1.yaml) to regenerate the CSV and the figure.
- References to the datasets are CSuite, ER/SF generators, as in Geffner et al.; SynTReN (Van den Bulcke et al., 2006); Sachs et al. (2005); IHDP (Hill, 2011); Twins (Almond et al., 2005).
- Use of generative AI: AI tools can be employed drafting, brainstorming and coding assistance – all members have to be aware of the content in this form and declare the use in the individual form.

15. Conclusion

In this project, there is a theoretical exploration of DECI that is based on Theorem 1 and the issue of causal graph recovery. The work covers such topics as representation and inference with the help of probabilistic graphical models (PGMs), factorization and conditional independence assumptions, guarantees of graph recovery, and performance metrics that are relevant for DECI. The present project is based on recent research in the field of causal inference, and besides the practical performance of DECI, one needs to identify theoretical conditions of its mathematical recovery guarantees.

The M2 milestone provides a technical basis for this theoretical study. It describes a DECI graphical model, learning and inference approaches, defines baseline and benchmark protocol and connects the mathematical analysis with an executable evaluation pipeline. Evaluation will include the impact of data condition changes on graph recovery and will be performed using PGM-relevant key performance indicators (KPIs).

Unfortunately, the provided materials of the project do not include any completed team-run KPI outputs. Thus, no empirical results can be reported at this stage. The next step is to run the described pipeline, save the outputs in the `Results/` folder, and update the experimental results and abstract. In M3, there will be more experiments and robustness testing to explore DECI performance and the relationship between its theory and practice.

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